Apêndice A
Tabelas

Identidades Trigonométricas

(1) $\text{sen}^2 x + \cos^2 x = 1$

(2) $1 + \text{tg}^2 x = \sec^2 x$

(3) $1 + \text{cotg}^2 x = \text{cosec}^2 x$

(4) $\text{sen}^2 x = 1/2(1 - \cos 2x)$

(5) $\cos^2 x = 1/2(1 + \cos 2x)$

(6) $\text{sen}\, 2x = 2\,\text{sen}\, x \cos x$

(7) $\text{sen}\, x \cos y = 1/2[\text{sen}(x - y) + \text{sen}(x + y)]$

(8) $\text{sen}\, x\, \text{sen}\, y = 1/2[\cos(x - y) - \cos(x + y)]$

(9) $\cos x \cos y = 1/2[\cos(x - y) + \cos(x + y)]$

Tabela de Derivadas

Nesta tabela u e v são funções deriváveis de x e c, α e a são constantes.

(1) $y = c \Rightarrow y' = 0$

(2) $y = x \Rightarrow y' = 1$

(3) $y = c \cdot u \Rightarrow y' = c \cdot u'$

(4) $y = u + v \Rightarrow y' = u' + v'$

(5) $y = u \cdot v \Rightarrow y' = u' \cdot v' + v \cdot u'$

(6) $y = \dfrac{u}{v} \Rightarrow y' = \dfrac{v \cdot u' - u \cdot v'}{v^2}$

(7) $y = u^\alpha\,(\alpha \neq 0) \Rightarrow y' = \alpha \cdot u^{\alpha - 1} \cdot u'$

(8) $y = a^u\,(a > 0, a \neq 1) \Rightarrow y' = a^u \cdot \ln a \cdot u'$

(9) $y = e^u \Rightarrow y' = e^u \cdot u'$

(10) $y = \log_a u \Rightarrow y' = \dfrac{u'}{u} \log_a e$

(11) $y = \ln u \Rightarrow y' = \dfrac{u'}{u}$

(12) $y = u^v \Rightarrow y' = v \cdot u^{v-1} \cdot u' + u^v \cdot \ln u \cdot v'\,(u > 0)$

(13) $y = \text{sen}\, u \Rightarrow y' = \cos u \cdot u'$

(14) $y = \cos u \Rightarrow y' = -\text{sen}\, u \cdot u'$

(15) $y = \text{tg}\,u \Rightarrow y' = -\sec^2 u \cdot u'$

(16) $y = \text{cotg}\,u \Rightarrow y' = -\text{cosec}^2 u \cdot u'$

(17) $y = \sec u \Rightarrow y' = \sec u \cdot \text{tg}\,u \cdot u'$

(18) $y = \text{cosec}\,u \Rightarrow y' = -\text{cosec}\,u \cdot \text{cotg}\,u \cdot u'$

(19) $y = \text{arc sen}\,u \Rightarrow y' = \dfrac{u'}{\sqrt{1-u^2}}$

(20) $y = \text{arc cos}\,u \Rightarrow y' = \dfrac{-u'}{\sqrt{1-u^2}}$

(21) $y = \text{arc tg}\,u \Rightarrow y' = \dfrac{u'}{1+u^2}$

(22) $y = \text{arc cotg}\,u \Rightarrow y' = \dfrac{-u'}{1+u^2}$

(23) $y = \text{arc sec}\,u, |u| \geq 1 \Rightarrow y' = \dfrac{u'}{|u|\sqrt{u^2-1}}, |u| > 1$

(24) $y = \text{arc cosec}\,u, |u| \geq 1 \Rightarrow y' = \dfrac{-u'}{|u|\sqrt{u^2-1}}, |u| > 1$

(25) $y = \text{senh}\,u \Rightarrow y' = \cosh u \cdot u'$

(26) $y = \cosh u \Rightarrow y' = \text{senh}\,u \cdot u'$

(27) $y = \text{tgh}\,u \Rightarrow y' = \text{sech}^2 u \cdot u'$

(28) $y = \text{cotgh}\,u \Rightarrow y' = -\text{cosech}^2 u \cdot u'$

(29) $y = \text{sech}\,u \Rightarrow y' = -\text{sech}\,u \cdot \text{tgh}\,u \cdot u'$

(30) $y = \text{cosech}\,u \Rightarrow y' = -\text{cosech}\,u \cdot \text{cotgh}\,u \cdot u'$

(31) $y = \text{argsenh}\,u \Rightarrow y' = \dfrac{u'}{\sqrt{u^2+1}}$

(32) $y = \text{argcosh}\,u \Rightarrow y' = \dfrac{u'}{\sqrt{u^2-1}}, u > 1$

(33) $y = \text{arg tgh}\,u \Rightarrow y' = \dfrac{u'}{1-u^2}, |u| < 1$

(34) $y = \text{arg cotgh}\,u \Rightarrow y' = \dfrac{u'}{1-u^2}, |u| > 1$

(35) $y = \text{arg sech}\,u \Rightarrow y' = \dfrac{-u'}{u\sqrt{1-u^2}}, 0 < u < 1$

(36) $y = \text{arg cosech}\,u \Rightarrow y' = \dfrac{-u'}{|u|\sqrt{1+u^2}}, u \neq 0.$

Tabela de Integrais

(1) $\int du = u + C$

(2) $\int \dfrac{du}{u} = \ln|u| + C$

(3) $\int u^\alpha\, du = \dfrac{u^{\alpha+1}}{\alpha+1} + C$ (α é constante $\neq -1$)

(4) $\int a^u\, du = \dfrac{a^u}{\ln a} + C$

(5) $\int e^u\, du = e^u + C$

(6) $\int \text{sen}\,u\, du = -\cos u + C$

(7) $\int \cos u\, du = \text{sen}\,u + C$

(8) $\int \text{tg}\,u\, du = \ln|\sec u| + C$

(9) $\int \text{cotg } u \, du = \ln |\text{sen } u| + C$

(10) $\int \text{cosec } u \, du = \ln |\text{cosec } u - \text{cotg } u| + C$

(11) $\int \sec u \, du = \ln |\sec u + \text{tg } u| + C$

(12) $\int \sec^2 u \, du = \text{tg } u + C$

(13) $\int \text{cosec}^2 u \, du = -\text{cotg } u + C$

(14) $\int \sec u \cdot \text{tg } u \, du = \sec u + C$

(15) $\int \text{cosec } u \cdot \text{cotg } u \, du = -\text{cosec } u + C$

(16) $\int \frac{du}{\sqrt{a^2 - u^2}} = \text{arcsen} \frac{u}{a} + C$

(17) $\int \frac{du}{a^2 + u^2} = \frac{1}{a} \text{arc tg} \frac{u}{a} + C$

(18) $\int \frac{du}{u\sqrt{u^2 - a^2}} = \frac{1}{a} \text{arc sec} \left|\frac{u}{a}\right| + C$

(19) $\int \text{senh } u \, du = \cosh u + C$

(20) $\int \cosh u \, du = \text{senh } u + C$

(21) $\int \text{sech}^2 u \, du = \text{tgh } u + C$

(22) $\int \text{cosech}^2 u \, du = -\text{cotgh } u + C$

(23) $\int \text{sech } u \cdot \text{tgh } u \, du = -\text{sech } u + C$

(24) $\int \text{cosech } u \cdot \text{cotgh } u \, du = -\text{cosech } u + C$

(25) $\int \frac{du}{\sqrt{u^2 \pm a^2}} = \ln |u + \sqrt{u^2 \pm a^2}| + C$

(26) $\int \frac{du}{a^2 - u^2} = \frac{1}{2a} \ln \left|\frac{u + a}{u - a}\right| + C$

(27) $\int \frac{du}{u\sqrt{a^2 \pm u^2}} = -\frac{1}{a} \ln \left|\frac{a + \sqrt{a^2 \pm u^2}}{u}\right| + C$

Fórmulas de Recorrência

(1) $\int \text{sen}^n u \, du = -\frac{1}{n} \text{sen}^{n-1} u \cos u + \frac{n-1}{n} \int \text{sen}^{n-2} u \, du$

(2) $\int \cos^n u \, du = \frac{1}{n} \cos^{n-1} u \text{ sen } u + \frac{n-1}{n} \int \cos^{n-2} u \, du$

(3) $\int \text{tg}^n u \, du = \frac{1}{n-1} \text{tg}^{n-1} u - \int \text{tg}^{n-2} u \, du$

(4) $\int \text{cotg}^n u \, du = -\frac{1}{n-1} \text{cotg}^{n-1} u - \int \text{cotg}^{n-2} u \, du$

(5) $\sec^n u \, du = \frac{1}{n-1} \sec^{n-2} u \text{ tg } u + \frac{n-2}{n-1} \int \sec^{n-2} u \, du$

(6) $\displaystyle\int \operatorname{cosec}^n u\, du = -\frac{1}{n-1}\operatorname{cosec}^{n-2} u \operatorname{cotg} u + \frac{n-2}{n-1}\int \operatorname{cosec}^{n-2} u\, du$

(7) $\displaystyle\int \frac{du}{(u^2+a^2)^n} = \frac{u(u^2+a^2)^{1-n}}{2a^2(n-1)} + \frac{2n-3}{2a^2(n-1)}\int \frac{du}{(u^2+a^2)^{n-1}}$